

arrive at the reverse statistic, or the probability that a strategy is not sustainable, the former value simply has to be subtracted from 1. So a 40% chance of sustainability in turn implies a 60% chance of “retirement ruin.”

The mathematical formula attempts to relate three retirement risk factors—inflation, investment/sequence of returns, and longevity risks—into a summary risk measure called the probability of sustainability. It generalizes the deterministic calculation (which had no uncertainty) at the beginning of this chapter, from constant numbers to random numbers.

To use this formula, you need to have a number of handy input factors or numbers. First, you need the estimated halfway mark for your retirement in years. This is the median remaining lifespan, which is denoted in the formula by MRL. You can get this number from a longevity or mortality table. If you can’t find it, ask your doctor or insurance agent. In fact, my MRL is currently 40 years. This number doesn’t necessarily imply that I am only planning to live for 40 more years, but rather that half my current cohort of 40-year-olds will reach the age of 80, while half will not.

Another important input factor is your planned inflation-adjusted retirement spending rate. Again, this factor or number is usually denoted in percentage terms; for example, 5% or 6% or 7% of your retirement nest egg. The third and fourth inputs you need for computing the probability of sustainability are (1) the anticipated inflation-adjusted risk and (2) return from your investment portfolio. I denote the expected return by the label AM and the volatility by the label VOL.

So, to start, make sure you have these four numbers handy; for example, 40 years (for median remaining lifespan), 5% spending (for retirement spending rate), 10% (for the annual expected rate of return on investments), and 20% (for volatility of your investments).

The following mathematical definition takes these four input factors—MRL, SPENDING, AM, VOL—and maps them into two summary variables. The summary variables are denoted by “alpha” and “beta” and defined in Figure 8.3.